

A supply chain is a directed container, and composing two is a Zappa–Szép product: a computed worked example (with the inventory-inconsistency obstruction in $H^2(\text{Sk}_{\mathcal{C}}; \mathbb{Z}/n) \cong \mathbb{Z}/n$)

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Abstract

This is a `COMPUTED` worked example, staging for the applications turn of the Kodamai grant: it instantiates machinery already `PROVED` elsewhere in a new domain and commits nothing to the book. We do three things. *(A)* We present a minimal supply chain `Procure` → `manufacture` → `ship` as an explicit small category and write out its Ahman–Uustalu directed container $(S, P, o, \oplus)$, machine-checking the five directed-container laws. *(B)* We model two flows sharing a warehouse as a small category $\mathcal{W}_{n,\varepsilon}$ carrying a cyclic *lot-cursor* token τ of order n at the shared node; fixing the token-internal relabelings as the right factor $\mathcal{D}$, the pairwise Zappa–Szép criterion and the (G) -obstruction-is- H^2 theorem (both cited, proved) give

$$[\omega(\mathcal{W}_{n,\varepsilon})] = \varepsilon \quad \text{in} \quad H^2(\text{Sk}_{\mathcal{C}}; \mathbb{Z}/n) \cong \mathbb{Z}/n,$$

so the joint chain serializes into a Zappa–Szép product $\mathcal{C} \bowtie \mathcal{D}$ *iff* $\varepsilon = 0$, i.e. iff the two routes to the same delivered good agree on provenance. Here $\varepsilon \in \mathbb{Z}/n$ is the *quantitative unit-mismatch* between the routes; the parity case $n = 2$ reproduces the proved orchestration re-entrancy bit. *(C)* The same machine, $n = 2$, is the merge of two ologs sharing a type: a naming-convention disagreement is the nonzero class. Every numerical claim is machine-checked in `scratch/supply_chain_zs.py`. **Grade discipline:** the general theorems are cited at `PROVED/LEAN-VERIFIED`; the domain instantiation is `COMPUTED`; the object-level claim “a real supply chain *is* this category” stays **open** (SEED Q4) — we build a faithful abstraction, not a fidelity claim.

1 Scope, grades, and cited scaffolding

Nothing here is new mathematics. The contribution is a *faithful minimal abstraction* of a supply chain in which the already-built container/Zappa–Szép machinery computes a meaningful invariant — exactly the move that promoted orchestration=ZS from analogy to computed to proved, now run in the supply-chain / olog domain. Four facts are used as black boxes.

- (T1) *DCont* $\simeq$ *SmallCat* (Ahman–Uustalu; our `DContCat.lean`, sorry-free, registry **equivalence-chain**, `LEAN-VERIFIED`). A directed container is exactly a small category; anything presenting a small category is a directed container for free.
- (T2) *Pairwise Zappa–Szép criterion* [1] (`PROVED`). For a small category $\mathcal{K}$ and wide $\mathcal{D} \subseteq \mathcal{K}$, there is a wide $\mathcal{C}$ with $(\mathcal{C}, \mathcal{D})$ a strict factorization system — equivalently a distributive law $\lambda : \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D}$, equivalently $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ — iff (L) each $\text{Hom}_{\mathcal{K}}(-, b)$ is a free right $\mathcal{D}$ -set and (G) free bases close into a wide subcategory.

- (T3) *The (G)-obstruction is an H^2 class* [2] (PROVED). Under (L) and hypothesis (H) (abelian vertex groups, trivial left action), $(G) \Leftrightarrow [\omega] = 0 \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$, with $\text{Sk}_{\mathcal{C}}$ the orbit category and $\mathcal{D}$ the vertex-group presheaf. Cohomology is Baues–Wirsching [5] / Rosebrugh–Wood [4], cited not reproved.
- (T4) *Orchestration re-entrancy* [3] (PROVED; $[\omega] = \varepsilon$ core LEAN-VERIFIED, `Reentrancy.lean`). The $n = 2$ instance of §3 below; we cite it as the parity anchor.

Ologs are Spivak’s prior art [6]; the knowledge-graph-as-copresheaf reading is folklore categorical databases. We import definitions, we do not reinvent them.

The one honest caveat. Whether a *real* supply chain genuinely presents a category — operations compose associatively, and the inventory-consistency path-equations hold on the nose — is SEED Open Question 4 and stays **open**. Everything below is a chosen minimal abstraction: the theorems apply to *it*, and the value is that they compute the right qualitative invariant. We never write “a supply chain *is* $\mathcal{W}_{n,\varepsilon}$.”

2 (A) The base chain as a directed container

Definition 1 (The procurement chain $\mathcal{C}_0$). Let $\mathcal{C}_0$ be the poset category on stages $\text{Src} < \text{Mfg} < \text{Dst}$ (supplier, manufacturing, delivery): objects are the three stages, and there is a unique morphism $i \rightarrow j$ whenever $i \leq j$, read as the *net operation* carrying material from stage i to stage j . Composition is do-then-do; thinness encodes “all routings from i to j that respect the order deliver the same material.”

By (T1), $\mathcal{C}_0$ is a directed container $(S, P, o, \oplus)$ (Ahman–Uustalu; our `DContCat.lean` dictionary):

$$\begin{array}{ll}
S = \{\text{Src}, \text{Mfg}, \text{Dst}\} & (\text{shapes} = \text{stages}), \\
P(s) = \{\text{operations out of } s\} = \{(s \rightarrow j) : j \geq s\} & (\text{positions} = \text{outgoing operations}), \\
o_s = (s \rightarrow s) = \text{id}_s & (\text{root} = \text{stay put}), \\
s \downarrow p = \text{cod}(p) & (\text{sub-shape} = \text{destination stage}), \\
p \oplus q = q \circ p & (\text{shift} = \text{do } p \text{ then } q).
\end{array}$$

Explicitly $P(\text{Src}) = \{\text{id}, \text{Src} \rightarrow \text{Mfg}, \text{Src} \rightarrow \text{Dst}\}$, $P(\text{Mfg}) = \{\text{id}, \text{Mfg} \rightarrow \text{Dst}\}$, $P(\text{Dst}) = \{\text{id}\}$, and e.g. $(\text{Src} \rightarrow \text{Mfg}) \oplus (\text{Mfg} \rightarrow \text{Dst}) = (\text{Src} \rightarrow \text{Dst})$: procure-then-ship *is* the direct delivery operation.

Proposition 2 (D-laws). $(S, P, o, \downarrow, \oplus)$ satisfies the five directed-container laws: D1 $s \downarrow o_s = s$; D2 $o_s \oplus p = p$; D3 $p \oplus o_{s \downarrow p} = p$; D4 $s \downarrow (p \oplus q) = (s \downarrow p) \downarrow q$; D5 $(p \oplus q) \oplus r = p \oplus (q \oplus r)$.

Proof. $\mathcal{C}_0$ is a thin poset category, so D1–D3 are the unit laws (identities are the $(s \rightarrow s)$), D4 says the destination of a composite is the destination of its second leg, and D5 is associativity of composition, all automatic in a poset. Machine-checked exhaustively over all composable p, q, r in `supply_chain_zs.py` part (A). $\square$

This is the object-level content the orchestration note omitted (it worked only at the category level): a supply chain, once abstracted to a category, is literally a polynomial comonoid / directed container, so the entire structure theory — the four monoidal products, the (co)monoid classifications, the chain rule — is available to it. That is a definitional dividend of (T1), graded COMPUTED (an instantiation), and it is *greenfield in our corpus*.

3 (B) Two flows sharing a warehouse: the family $\mathcal{W}_{n,\varepsilon}$

Now the composition story. Two flows meet at a warehouse Wh . The warehouse keeps an internal *lot-cursor*: a cyclic pointer τ of order n into n interchangeable bins / lot-slots (FIFO position mod n , or a bin index). Physically τ moves nothing — it is pure bookkeeping — but a unit leaving the warehouse carries whatever cursor value was current. Advancing the cursor is an automorphism τ with $\tau^n = \text{id}_{\text{Wh}}$, so $\text{End}(\text{Wh}) \cong \mathbb{Z}/n$.

Definition 3 (The warehouse family $\mathcal{W}_{n,\varepsilon}$). For $n \geq 1$ and $\varepsilon \in \mathbb{Z}/n$, let $\mathcal{W}_{n,\varepsilon}$ be the small category with objects Wh (warehouse), Pr (processing/manufacture), De (delivery), and

$$\begin{aligned} \text{End}(\text{Wh}) &= \{\tau^k : k \in \mathbb{Z}/n\} \cong \mathbb{Z}/n, & \text{End}(\text{Pr}) &= \{\text{id}\}, & \text{End}(\text{De}) &= \{\text{id}\}, \\ \text{Hom}(\text{Wh}, \text{Pr}) &= \{p\tau^k : k \in \mathbb{Z}/n\}, & \text{Hom}(\text{Wh}, \text{De}) &= \{q\tau^k : k \in \mathbb{Z}/n\}, & \text{Hom}(\text{Pr}, \text{De}) &= \{s, s_2\}, \end{aligned}$$

no arrows into Wh except its endomorphisms, and no arrows $\text{De} \rightarrow \cdot$ except id_{De} (acyclic: $\text{Wh} \rightarrow \text{Pr} \rightarrow \text{De}$ with a direct $\text{Wh} \rightarrow \text{De}$). Here $p\tau^k := p \circ \tau^k$, all distinct; the two processing operations compose through the middle by

$$s \circ p = q, \quad s_2 \circ p = q \cdot \tau^\varepsilon,$$

extended by right τ -equivariance ($s \circ p\tau^k = q\tau^k$, $s_2 \circ p\tau^k = q\tau^{k+\varepsilon}$).

Reading. p dispatches a lot to processing; q is the direct/expected delivery recorded by the warehouse's books; s is the nominal production-and-ship line and s_2 is an alternative route (rework, a second line, an expedited lane) that delivers the *same good*. The single datum ε records how many lot-positions s_2 *advances the provenance cursor* relative to the nominal route: $\varepsilon = 0$ means the rework line preserves lot-provenance, $\varepsilon \neq 0$ means it shifts it by ε units. The wide right factor of bookkeeping moves is

$$\mathcal{D} = \{\tau^k : k \in \mathbb{Z}/n\} \cup \{\text{id}_{\text{Pr}}, \text{id}_{\text{De}}\} \subseteq \mathcal{W}_{n,\varepsilon},$$

the “warehouse-internal relabeling” subcategory — the *second chain* that shares the warehouse with the physical-flow chain.

Lemma 4 ($\mathcal{W}_{n,\varepsilon}$ is a category). *For every n, ε , $\mathcal{W}_{n,\varepsilon}$ is a well-defined small category.*

Proof. Only two families of composites need specifying: the $\mathbb{Z}/n$ -action of τ on Wh -sourced arrows (a group action, $\tau^n = \text{id}$), and the two length-two composites $s \circ p, s_2 \circ p : \text{Wh} \rightarrow \text{De}$. All longer composites factor through the sink De or start with a Wh -endomorphism, so associativity reduces to right-equivariance, $(f \circ p) \circ \tau^k = f \circ (p\tau^k)$, which holds by definition of $f \circ p\tau^k$. Machine-checked (full associativity table) for eight (n, ε) pairs in `supply_chain_zs.py` part (B). $\square$

Lemma 5 ((L) and (H) hold, uniformly in ε). *Every hom-presheaf $\text{Hom}_{\mathcal{W}_{n,\varepsilon}}(-, b)$ is a free right $\mathcal{D}$ -set, and $(\mathcal{W}_{n,\varepsilon}, \mathcal{D})$ satisfies hypothesis (H).*

Proof. (L). The right $\mathcal{D}$ -action is precomposition; it is trivial on Pr - and De -sourced arrows and is the regular $\mathbb{Z}/n$ -action on Wh -sourced arrows. Orbit decomposition: $\text{Hom}(-, \text{Wh}) = \{\tau^k\}$ (one regular $\mathbb{Z}/n$ -orbit); $\text{Hom}(-, \text{Pr}) = \{p\tau^k\} \sqcup \{\text{id}_{\text{Pr}}\}$; $\text{Hom}(-, \text{De}) = \{q\tau^k\} \sqcup \{s\} \sqcup \{s_2\} \sqcup \{\text{id}_{\text{De}}\}$. Each Wh -apex orbit is a free $\mathbb{Z}/n$ -torsor (the τ^k are distinct), each other orbit a singleton with trivial group; so every hom-presheaf is a coproduct of representable $\mathcal{D}$ -sets, i.e. free. This uses only distinctness of the $p\tau^k$ and $q\tau^k$; it is independent of ε (moving $s_2 \circ p$ inside the orbit $\{q\tau^k\}$ does not change orbits). (H). $\mathcal{D}$ is a disjoint union of abelian vertex groups $G_{\text{Wh}} \cong \mathbb{Z}/n$, $G_{\text{Pr}} = G_{\text{De}} = 0$;

the only nontrivial vertex group sits at $\mathbf{Wh}$, which is a *source* (no generator targets $\mathbf{Wh}$), so the left action (postcomposition by a target automorphism) is trivial. Machine-checked (freeness of every orbit) in part (B). $\square$

Proposition 6 (Orbit category, complex, and $H^2 \cong \mathbb{Z}/n$). *$\mathbf{Sk}_C$ has objects $\mathbf{Wh}, \mathbf{Pr}, \mathbf{De}$ and non-identity generators $[p] : \mathbf{Wh} \rightarrow \mathbf{Pr}$, $[q] : \mathbf{Wh} \rightarrow \mathbf{De}$, $[s], [s_2] : \mathbf{Pr} \rightarrow \mathbf{De}$, with $[s] * [p] = [s_2] * [p] = [q]$ independently of ε (because q and $q\tau^\varepsilon$ lie in one orbit). The presheaf has $\mathcal{D}(\mathbf{Wh}) = \mathbb{Z}/n$, $\mathcal{D}(\mathbf{Pr}) = \mathcal{D}(\mathbf{De}) = 0$, and all restriction maps vanish (no generator targets $\mathbf{Wh}$). Hence $C^1 \cong \mathbb{Z}/n^2$ (coords $h[p], h[q]$), $C^2 \cong \mathbb{Z}/n^2$ (coords $\omega([s], [p]), \omega([s_2], [p])$), $C^3 = 0$, and*

$$(\delta^1 h)([s], [p]) = (\delta^1 h)([s_2], [p]) = h[p] - h[q],$$

so $B^2 = \{(t, t) : t \in \mathbb{Z}/n\}$ is the diagonal and

$$H^2(\mathbf{Sk}_C; \mathcal{D}) = \mathbb{Z}/n^2 / \langle (1, 1) \rangle \cong \mathbb{Z}/n, \quad [(a, b)] \mapsto b - a.$$

Proof. Identical bookkeeping to [3, §4] with $\mathbb{Z}/n$ in place of $\mathbb{Z}/2$: the generators, the vanishing restrictions, and the two composable pairs $([s], [p])$, $([s_2], [p])$ are exactly as there, and the Baues–Wirsching coboundary formula $(\delta^1 h)(o_2, o_1) = \varphi_{o_1}(h(o_2)) - h(o_2 \circ o_1) + h(o_1)$ reduces to $h[p] - h[q]$ since $\varphi_{[p]} = \mathcal{D}([p]) : 0 \rightarrow \mathbb{Z}/n$. Machine-checked ($B^2 = \text{diagonal}$) for all tested n in part (B). $\square$

Theorem 7 (The class is the unit-mismatch). *With the natural transversal $T = \{p, q, s, s_2\}$, the defect 2-cocycle is $\omega_T = (0, \varepsilon) \in \mathbb{Z}/n^2$, so*

$$[\omega(\mathcal{W}_{n,\varepsilon})] = \varepsilon \in H^2(\mathbf{Sk}_C; \mathcal{D}) \cong \mathbb{Z}/n.$$

Proof. Both composable pairs have target orbit $\text{orb}(q)$ with chosen generator q . From $s \circ p = q = q \circ \text{id}_{\mathbf{Wh}}$ we read $\omega_T([s], [p]) = 0$, and from $s_2 \circ p = q \cdot \tau^\varepsilon = q \circ \tau^\varepsilon$ we read $\omega_T([s_2], [p]) = \tau^\varepsilon = \varepsilon$. So $\omega_T = (0, \varepsilon)$, and its class under $[(a, b)] \mapsto b - a$ is ε . Machine-verified $\omega_T = (0, \varepsilon)$ and $[\omega] = \varepsilon$ in part (B). $\square$

Corollary 8 (Inventory-consistency dichotomy). *For the warehouse abstraction $\mathcal{W}_{n,\varepsilon}$ with bookkeeping right factor $\mathcal{D}$:*

$$\mathcal{W}_{n,\varepsilon} = \mathcal{C} \bowtie \mathcal{D} \text{ exists} \iff \varepsilon = 0 \iff \text{both routes to De agree on lot-provenance.}$$

When $\varepsilon \neq 0$ the obstruction $[\omega] = \varepsilon$ is a nonzero element of $H^2 \cong \mathbb{Z}/n$: a genuine inventory inconsistency, the two routes to the same delivered good disagreeing by ε lot-positions, and no serialization of the physical flow with the warehouse bookkeeping exists.

Proof. (L) holds (Lemma 5) so by (T2) the product exists iff (G); (H) holds so by (T3) (G) iff $[\omega] = 0$; by Theorem 7 $[\omega] = \varepsilon$. Chain the equivalences. As an independent (non-cohomological) cross-check, part (B) brute-forces the strict factorization systems: $\#\text{SFS} = n$ when $\varepsilon = 0$ (a $\mathbb{Z}/n$ -torsor of ways to choose the canonical lot-labeling) and $\#\text{SFS} = 0$ when $\varepsilon \neq 0$. $\square$

Remark 9 (Why $\mathbb{Z}/n$, not just parity). *Orchestration [3] needed only $\mathbb{Z}/2$: a turn token is one bit. Inventory is naturally a counter, and the obstruction lands in the counter’s group: the class $[\omega] = \varepsilon$ is not just “inconsistent yes/no” but *by how many units* the two provenance routes disagree, valued in $\mathbb{Z}/n$. The proved parity theorem is the $n = 2$ face of this. The gauge freedom $B^2 = \langle (1, 1) \rangle$ is exactly the liberty to declare which lot-slot is “slot zero”: it shifts *both* routes together, so only their *difference* ε is observable — the categorical avatar of a flat $\mathbb{Z}/n$ -bundle being trivial iff its holonomy (here the provenance drift) vanishes.*

4 (C) Olog sibling: merging two schemas along a shared type

The same abstraction serves ontology/knowledge-graph merges. The olog $\text{Book} \xrightarrow{\text{has}} \text{Author} \xrightarrow{\text{has}} \text{Name}$ is the free category on that graph; by (T1) it is a directed container with shapes = types, positions out of a type = the aspects available from it, $\oplus$ = aspect composition, path-equations = facts (D-laws machine-checked, part (C)). Merging it with a second olog sharing Author (e.g. $\text{Author} \rightarrow \text{Affiliation}$) is composing two directed containers over the shared type — a Zappa–Szép problem. Take $\text{Wh} := \text{Author}$ carrying a $\mathbb{Z}/2$ token = choice of naming convention (“Last, First” vs. “First Last”), $\text{Pr} := \text{name-record}$, $\text{De} := \text{rendered display}$. Then Corollary 8 at $n = 2$ reads: the merge succeeds ($\mathcal{C} \bowtie \mathcal{D}$ exists) iff the two schemas agree on the convention ($\varepsilon = 0$); a disagreement is the nonzero class $[\omega] = 1$, a *merge conflict* — the same H^2 bit as supply-chain inconsistency and agent re-entrancy. Machine-checked, part (C).

5 Status, regimes, and grant relevance

The computed table.

regime	$[\omega] \in \mathbb{Z}/n$	outcome	meaning
$\varepsilon = 0$ (any n)	0	composes, $\#\text{SFS} = n$	routes agree on provenance
$\varepsilon \neq 0$ (any n)	ε	obstructed, $\#\text{SFS} = 0$	inventory off by ε units
$n = 2, \varepsilon = 1$	1	obstructed	= orchestration re-entrancy bit [3]
olog $n = 2$	0 / 1	merges / conflicts	schema conventions agree / clash

Eight (n, ε) instances verified in `scratch/supply_chain_zs.py` (category axioms, (L), $\text{Sk}_{\mathcal{C}}$ shape, $B^2 = \text{diagonal}$, $\omega_T = (0, \varepsilon)$, $[\omega] = \varepsilon$, and the $\#\text{SFS}$ cross-check).

What is proved vs. computed. The general theorems (T1)–(T4) are PROVED/LEAN-VERIFIED and are cited, not re-derived. The *domain instantiation* — that this abstraction of supply chains and olog merges computes the inventory-inconsistency / merge-conflict class — is COMPUTED illustration. The object-level modelling claim “a real supply chain presents a category” is **open** (SEED Q4); we have exhibited a faithful minimal abstraction, not asserted fidelity, exactly as the orchestration note did for agents.

Grant line. Supply-chain composability, ontology-merge consistency, and agent-orchestration re-entrancy are the *same degree-two obstruction* on the handoff category, and in the supply-chain instance the class is refined from a bit to a $\mathbb{Z}/n$ -valued *unit-count* of the discrepancy. This is the Path-5 Impact spine: the applications are not new mathematics but new *instances of one obstruction theorem*, and that is the honest strength of the story.

References

- [1] MacBeth, *The pairwise Zappa–Szép criterion*, Kodamai note, 10 June 2026.
- [2] MacBeth, *The global-closure obstruction (G) is a cohomology class*, Kodamai note, 11 June 2026.

- [3] MacBeth, *The re-entrancy obstruction is the token-mutation bit: an analytic proof for orchestration Zappa–Szép products*, Kodamai note, 20 July 2026 ($[\omega] = \varepsilon$ core Lean-verified, `Reentrancy.lean`).
- [4] R. Rosebrugh, R. J. Wood, *Distributive laws and factorization*, J. Pure Appl. Algebra 175 (2002), 327–353.
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- [6] D. I. Spivak, R. E. Kent, *Ologs: a categorical framework for knowledge representation*, PLoS ONE 7(1) (2012), e24274.